Image Processing

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Homework 1

You are not allowed to use any ready-made image-processing functions. You must perform all operations manually using NumPy and … and your own mathematical implementations.
THANK YOU FOR ATTENTION TO THIS MATTER.

Q1-Choose a **colorful RGB image (3 channels)** and do the following steps:

1.Load the RGB image in Python. (Using `from PIL import Image`)
2.Convert the image into: a. a **Grayscale** image b. a **Binary (black and white)** image.
3.Display all three images: the original RGB, the Grayscale version, and the Binary version.
4.Print the array values and the **shape** of each image and compare them.

Q2 – Manual Image Resizing (Downsampling by Averaging)

After converting the image to grayscale, perform the following steps:

1.Downsample the image to **1/2 size** by dividing the image into **2×2 blocks** and replacing each block with the **average of its pixel values**.

2.Downsample the image to **1/4 size** by dividing the image into **4×4 blocks** and replacing each block with the **average of the pixels in each block**.

3.Constraint: You are strictly forbidden from using any ready-made resize or scaling functions (such as cv2.resize, PIL.resize, etc.). The downsampling must be implemented manually using **NumPy operations or loops**.

4.Display the original grayscale image, the 1/2 scale image, and the 1/4 scale image, and compare their shapes.

Q3-Negative of a Binary Image:

Choose a picture and make it **binary (black and white) image** where pixel values are only **0** (black) or **255** (white), perform the following:

1.Load the binary image using Pillow and convert it to a NumPy array.
2.Create the **negative** of the image **manually**, using only NumPy.
   1. Black (0) must become White (255)
   2. White (255) must become Black (0)
   3. You are not allowed to use any ready-made image-processing functions.
3.Display both the original binary image and its negative version.
4.Print and compare their array values.

Q4-Gamma Correction Application

A grayscale image is given as a NumPy array with pixel values in the range **0–255**.

1.Explain what **Gamma Correction** is and why it is used in image processing.

2.Write the mathematical formula used for Gamma Correction.

3.Using **NumPy only**, implement Gamma Correction for a grayscale image with a chosen gamma value (for example $\gamma=0.5$ and $\gamma=2.0$).

4.Display the **original image** and the **gamma-corrected images** for the two gamma values.

5.Briefly explain how changing the gamma value affects the brightness of the image.

Q5-Mean Filtering (3×3 Smoothing)

1.Explain the purpose of **smoothing (low-pass) filters** in the spatial domain.

2.Write the 3×3 averaging mask and explain how it operates on the image.

3.Implement 3×3 mean filtering **manually**:

1. Use zero-padding or replicate-padding for borders.
2. Use nested loops or convolution implemented by NumPy operations (but do not use any ready-made convolution functions).

4.Show the original and the smoothed images and discuss what happens to noise and edges.

Q6– Laplacian-Based Sharpening

1.Write the standard 3×3 Laplacian mask used for image sharpening.

2.Explain the idea of sharpening by **adding** a scaled Laplacian to the original image.

3.Implement the Laplacian filtering manually (no cv2.filter2D, no ready-made convolution).

4.Generate a sharpened version of the image and comment on the effect on edges and flat regions.

Q7 – Manual Histogram Equalization (No ready-made functions)

You are given a grayscale image and its intensity histogram $h(r_k)$ where r_k are gray levels:

1.Write the mathematical formula for the **CDF-based transformation** used in histogram equalization.

2.Explain, in one paragraph, how histogram equalization tends to improve the contrast of an image.

3.Implement **manual histogram equalization** using NumPy only:

1. Compute the normalized histogram.
2. Compute the cumulative distribution function (CDF).
3. Map old gray levels to new ones using the CDF.

4.Compare the original histogram and the equalized histogram and describe the difference.

Deliverables:

For each question, your submission must include:

- The relevant **mathematical formulation**
- A clear **conceptual explanation**
- The **Python implementation using NumPy**
- The **resulting output or images**